

C Programming Question Bank

Assignment 2

SECTION B: INTERMEDIATE LEVEL (Applied Programs)

B1. Write a program to check whether a given year is a leap year using nested `if` / logical operators.

B2. Write a program to find the largest among three numbers using `if-else if-else`.

B3. Write a program that reads a character and determines whether it is a vowel or a consonant.

B4. Write a program to calculate the grade of a student based on marks entered (using `else if` ladder):

- $\geq 90 \rightarrow A, \geq 75 \rightarrow B, \geq 60 \rightarrow C, < 60 \rightarrow \text{Fail}$

B5. Predict the output:

```
int a = 5, b = 2;
```

```
float c;
```

```
c = a / b;
```

```
printf("%f", c);
```

Explain *why* the output is not `2.5`, and how to fix it.

B6. Write a program to check whether a triangle is valid given its three angles (sum must equal 180, and each angle > 0).

B7. Explain operator precedence with an example involving `+`, `*`, and `()`. Evaluate:

```
int result = 10 + 2 * 3 - (4 / 2);
```

B8. Write a program to input a number and check if it is divisible by both 3 and 5 using logical `&&` operator.

B9. What is implicit type conversion? Predict the output:

```
char ch = 'A';  
  
int x = ch + 1;  
  
printf("%d %c", x, x);
```

B10. Write a program using nested **if-else** to categorize a person's BMI (Underweight/Normal/Overweight) based on input weight and height.

B11. Differentiate between **if-else** and **else if** ladder with a program example where **else if** is more efficient.

B12. Predict the output and explain the operator behavior:

```
int a = 5;  
  
printf("%d %d %d", a++, ++a, a);
```

B13. Write a program using nested **if** that first checks whether a number lies between 1 and 100, and if so, further classifies it as belonging to the range 1–50 or 51–100.

B14. Write a menu-driven program using **switch** to convert temperature between Celsius, Fahrenheit, and Kelvin, based on the user's choice of conversion (e.g., 1 = C to F, 2 = F to C, 3 = C to K).

B15. Write a program using **switch-case** to print the number of days in a given month (1–12). For month 2 (February), use a **nested if** inside the **switch** case to additionally check for a leap year.

B16. Compare nested **if-else** and **switch-case** statements. Give one example situation where **switch** is clearly the better choice, and one where it **cannot** be used at all even though **if-else** can (hint: think about the type of condition being checked, not just the number of cases).

B17. Write a program using nested **if** to determine ticket pricing at a cinema: first check if the person is a student (discount applies), then within that, check if the show is a weekday or weekend (different discount percentages for each combination).